Qing Yi Chen

PORTFOLIO

Aerospace Engineering MEng '27
Mechanical Engineering BS '27, Minor in Business
Cornell University '27 Junior

ABOUT ME

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Hello! My name is Qing and I'm a first-generation college junior studying Mechanical Engineering at Cornell University. I'm expected to graduate with a Bachelor of Science (BS) and minor in business in May 2027 and additionally expecting to graduate with a Master in Engineering (MEng) in Aerospace Engineering in December 2027.

I have an extensive background in technical engineering project team experience working with Hyperloop technology, thermoelectric and solar prototyping undergraduate research, an engineering asset management internship at the NYC DEP (Department of Environmental Protection), and a long history of peer & group mentorship.

WORK EXPERIENCE

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Experience

CORNELL HYPERLOOP PROJECT TEAM

– STRUCTURES SUBTEAM (2023-PRESENT)

NYC DEPARTMENT OF ENVIRONMENTAL PROTECTION

– ASSET MANAGEMENT SUMMER COLLEGE INTERN (SUMMER 2025)

Research

CO-GEN ZT RESEARCH GROUP

– THERMALJANSYS AND BODY STRUCTURES SUBTEAMS (2025-PRESENT)

Campus Activities

CORNELL ENGINEERING PEER ADVISOR (2024-PRESENT)

THERMODYNAMICS GRADER (2025-PRESENT)

INTRO TO MECHANICAL ENGINEERING GRADER (FALL 2024)

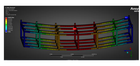
AMERICAN SOCIETY OF MECHANICAL ENGINEERS (2023-PRESENT)

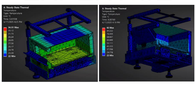
QSTEM (2023-PRESENT)

GUIDING EYES FOR THE BLIND (2023-PRESENT)

SKILLS & EXPERTISE

04





CAD, FEA, CFD, 3D Modeling

– SolidWorks
– Autodesk Fusion 360
– ANSYS
– OnShape

Coding Languages

– Python
– MATLAB
– Java

Technical

– 3D Printing
– Laser Cutting
– Drafting
– Prototyping
– Machining

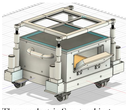
Other

– Microsoft Office (Excel, Power BI, Word, PowerPoint, etc.)
– Google Suite
– Confluence
– LaTeX

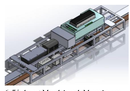
MY WORK

05

Air Cooled 100 cell Battery Pack & BMS Enclosure
CORNELL HYPERLOOP PROJECT TEAM
(SOLIDWORKS)



Thermoelectric Sun tracking Solar Furnace
CO-GEN ZT RESEARCH GROUP
(AUTODESK FUSION 360)

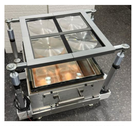


6.5ft long Machined Aluminum & PA6 Press fits Chassis
CORNELL HYPERLOOP PROJECT TEAM
(SOLIDWORKS)

06

TESTING MINI-HYPERLOOP POD TO SHOWCASE ELECTROMAGNETIC LEVIATION AND ACCELERATION CAPABILITIES

FABRICATING THE AERODYNAMIC CARBON FIBER AEROSHELL FOR THE FULL-SIZE HYPERLOOP POD USING CNC-CUT HIGH DENSITY FOAM, DURATEC, AND EPOXY CURE



IMPLEMENTING SUN-TRACKING FEATURES IN THE FOUR ACTUATORS OF THE THERMOELECTRIC FURNACE VIA AI ALGORITHM, CONCENTRATING/AMPLIFYING SUN RAYS ON THE COPPER COOKING PLATE VIA FRESNEL LENSES, AND PRODUCING HIGH VOLTAGE OUTPUT VIA INTERNAL TEGS

OTHER PROJECTS

A SAMPLE OF PROJECTS I'VE PRODUCED FOR OTHER ENGINEERING COURSES

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MAE 3280 Final Group Work: Exploring a System of Interest

Report

Studies: Solids 101

Author: Qing Yi Chen

Topic of interest: Aerodynamic Glider (Hollow-core) using balsa supports and PLA connector

Motivation: This project models the relative importance of four aerodynamic variables (supporting wing design) in determining the maximum distance a glider can travel before it touches the ground. The goal is to understand how changes in wingspan, wing angle, and support shape affect overall performance, and to identify which design choices have the greatest impact on flight distance. This work provides an overall view of the design trade-offs between aerodynamic efficiency, structural support, and material usage, which is important for practical glider design and optimization.

Assumptions:
- Wing angle is measured relative to the horizontal plane
- Connector length is fixed at 0.5 in
- Connector length is not a design variable
- Wingspan is measured from tip to tip
- Wing thickness is fixed at 0.25 in
- Wing is modeled as a flat plate with no airfoil shape
- Support shape is modeled as a flat plate with no airfoil shape
- Drag coefficients are assumed constant and independent of Reynolds number
- Lift and drag forces act through the center of pressure
- Gravity acts through the center of mass
- Air density and kinematic viscosity are assumed constant at standard conditions
- Ground is modeled as a flat, rigid surface
- Collision with the ground is treated as a stop condition
- Glide is assumed to occur in still air (no wind)
- Initial velocity is assumed to be zero
- Rotational effects are neglected
- Stability and control effects are not considered

Table 1: Variable Ranges

Variable	Lower Bound	Upper Bound
Wingspan (in)	12	24
Wing Angle (°)	0	15
Support Shape	Triangular Prism	Rectangular Prism

Table 2: Fixed Parameters

Parameter	Value
Connector Length (in)	0.5
Wing Thickness (in)	0.25
Wing Shape	Flat plate, no airfoil
Support Shape	Flat plate, no airfoil
Drag Coefficient	Constant, independent of Reynolds number
Lift Coefficient	Constant, independent of Reynolds number
Gravity	9.81 m/s²
Air Density	1.225 kg/m³
Kinematic Viscosity	1.5 × 10⁻⁵ m²/s
Ground Model	Flat, rigid surface
Collision Model	Stop condition
Wind	Still air (no wind)
Initial Velocity	0 m/s
Rotational Effects	Neglected
Stability and Control	Not considered

Table 3: MAE 3280 examples of which used to this group work

Year	Class
2025	MAE 3280

Aerodynamic Glider

36 INCH-LONG SELF-MADE BALSA & SPRUCE WOOD GLIDER DESIGNED, CALCULATED, AND TESTED TO FLY 20+ METERS

(INTRO TO AERONAUTICS)

Modular Shower Caddy

DESIGNED, 3D PRINTED, AND LASER CUT FROM CUSTOMER REQUEST

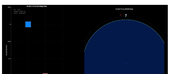
(INTRO TO MECHANICAL DESIGN)

Autonomous Orbital Rendezvous

MODELS THE DYNAMICS OF A CHASER SATELLITE APPROACHING A TARGET SATELLITE USING THE CLOHESSY-WILTSHIRE EQUATIONS IN STATE SPACE FORM AND DEMONSTRATES PD CONTROL OF A MIMO SYSTEM.

(SYSTEM DYNAMICS)

08



Qing Yi Chen

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LinkedIn Website CO-Email: College Junior
New York, NY
Willing to Relocate

EDUCATION

Cornell University Expected Graduate: 2027
Master of Engineering, Aerospace Engineering Ithaca, NY

Cornell University Expected May 2027
B.S. in Mechanical Engineering, Mechanical Engineering, Electro Business Minor (In Sequence (EBME)) Ithaca, NY
- Relevant Coursework: Thermodynamics, Mechanics of Materials, Thermodynamics, Statics, Intro to Aerodynamics, System Dynamics, Mechanical Design, Theory & Quantum Physics, Electricity & Magnetism Physics, Intro Science Computing, Intro to Aerospace Structures, Intro to Mechanical Engineers (MEM), gEDC, Finding Gyro for the Blind

EXPERIENCE

New York City Department of Environmental Protection (NYC DEP) June 2025 - August 2025
Summer College Intern, Asset Management Division Flushing, NY
- Collaborated with enterprise Systems to design consequence of failure (COF) scores in EAM, assets and condition of assets, and risk scores for over 1,000 assets, ensuring data accuracy and alignment with business requirements
- Utilized Excel and Power BI for large-scale data analysis & management, dashboard creation, and data visualization, supporting strategic decision-making and contributing to a model for over 1,000 EAM assets and their condition

Cornell Hyperloop Project Team October 2023 - Present
Structures Subteam Member Ithaca, NY
- Designed, 3D printed, and manufactured an air-cooled battery pack enclosure to hold the moving pod's 100 lithium-ion cells
- Develop a co-0 fitting conduit aluminum chassis, reducing its weight by 50% from previous carbon steel conduit
- Design, 3D print, and manufacture a 6.5ft long aluminum chassis to fit the battery pack and other components
- CAD model, fabricate, and install custom 3D-printed mounts for microcontrollers, sensors, and electrical wiring

Co-Gen Sun ZT Research Group January 2025 - Present
Thermoelectric Sun Tracking Solar Furnace Lead Mechanical Engineer, Thermal-AI/ML Subteam Ithaca, NY
- Coordinate a sustainable co-cooking thermoelectric furnace capable of generating and storing solar electricity using thermoelectric generators (TEGs) and a sun-tracking concentrator, integrating mechanical design, electronics, and AI-based control
- Improve TEG conversion significantly by 25%, from overall voltage output, upgrade probability, and reverse polarity, using Nanofluids Cross MOC-LSTM, and engineer our own experimental dual-axis prototype energy harvesters for the lab

Cornell Thermodynamics Course August 2025 - Present
Thermodynamics Grader Ithaca, NY
- Responsible for grading assignments and quizzes for 250 students and lead Brainstorm recitations in the course, Thermo
- Substituted course material on heating, cooling and power cycles, refrigeration, heat pumps, energy, entropy, etc.

Cornell Introduction to Mechanical Engineering Course August 2024 - December 2024
Intro to Mechanical Engineering Grader Ithaca, NY
- Evaluated assignments and exams, ensuring fair and consistent grading based on conceptual and analytical understanding
- Strengthened engineering concepts material covering kinetics, kinematics, power, stress, strain, energy, torque, Mohr's circle, etc.

Cornell Society of Women Engineers November 2024 - Present
Outreach Committee Member Ithaca, NY
- Lead and coordinate outreach and admin projects, networking events, resume reviews, and career-related workshops

SKILLS & LANGUAGES
Software: Microsoft Office, LaTeX, Facebook Ads, MATLAB, Solidworks, Ansys, Computational Fluid Dynamics (CFD), Finite Element Analysis (FEA), 3D Modeling, Laser Cutting, Machining, Microsoft Office, Google Suite, Confluence
Languages: Fluent English, conversational Taiwanese, elementary Mandarin

Qing Yi Chen

RESUME

09

File Link:
https://drive.google.com/file/d/1g6VUcSGQKW4D1FTokFAJCDH_hhRff600/view?usp=sharing

Qing Yi Chen

LET'S CONNECT

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